Use Case: Tic Tac Toe Home Menu

Iteration: 1

Primary Actor: Player

Goal In Context: To enable the player to access the main game interface for Tic Tac Toe and select options such as starting a new game, viewing the leaderboard, or joining the matchmaking queue.

Preconditions:

- The player is logged in and has access to their profile.
- The Tic Tac Toe module is active.

Trigger: The player navigates to the Tic Tac Toe game section from the Player Profile Home Menu.

Scenario:

1. The player clicks “Tic Tac Toe” from the Player Profile Home Menu.
2. The system loads Tic Tac Toe Home Menu.
3. Player is given options to either view the Tic Tac Toe leaderboard or join the matchmaking queue.
4. The player selects their desired option (e.g., “Join Queue”).
5. The system directs the player to the chosen option.

Post Conditions: The player is successfully able to access their chosen feature in the Tic Tac module.

Exceptions:

- **E1:** The Tic Tac Toe module is unavailable. The system shows an error message and redirects the player to the Player Profile Home Menu.

Priority: High

When Available: When player is logged in & navigates to Tic Tac Toe

Frequency of Use: Frequent, as this is the starting point for Tic Tac Toe-related activities.

Channel to Actor: User Interface (UI)

Secondary Actors: None

Channels to Secondary Actors: N/A

Open Issues: N/A

Use Case: Tic Tac Toe Game Queue

Iteration: 1

Primary Actor: Player

Goal In Context: To allow the player to enter a matchmaking queue to find an opponent for the Tic Tac Toe game.

Preconditions:

- The player has accessed the Tic Tac Toe Home Menu.
- Matching making feature is working.

Trigger: The player selects the option to join the Tic Tac Toe game queue.

Scenario:

1. The player selects “Join Queue” from the Tic Tac Toe Home Menu.
2. The system adds the player to the Tic Tac Toe game queue.
3. The system searches for a opponent.
4. Once opponent is found, system notifies the player and initiates the Tic Tac Toe Game Instance.

Post Conditions: Player either joins a match or gets a message that a match is not currently available.

Exceptions:

- **E1:** No players are available in the queue, so the system displays a “No Match Found” message.
- **E2:** Matchmaking service is offline or is malfunctioning so the system shows an error message and returns the player to the Tic Tac Toe Home Menu.

Priority: High

When Available: When the player clicks to enter a queue.

Frequency of Use: Moderate, depending on matchmaking demand for users

Channel to Actor: UI notification

Secondary Actors: Matchmaking service

Channels to Secondary Actors: System-to-system communication with matchmaking service

Open Issues:

- Should players be matched based on certain skill level or expertise?
- Extension to above: If there isn't a player that can match skill levels, can the user be notified if & when a match is found?